

$$\theta = 37.5^\circ$$

**18 Ans: D**

$$d = 10^{-3} / 500 = 2.0 \times 10^{-6}$$

$$d \sin 45^\circ = n\lambda$$

$$(2.0 \times 10^{-6}) \sin 45^\circ = n(600 \times 10^{-9})$$

$$n = 2.36$$

Hence, max order = 2

$$\text{Max number of bright fringes} = 2 + 2 + 1 = 5$$

**19 Ans: B**